

Project Design Phase

Proposed Solution Template

Date	19 July 2026
Team ID	
Project Name	Nutrition Assistant
Team Members	Kulasri Durga Deepika Bora Polimera Girija(Team lead)
Maximum Marks	2 Marks

Proposed Solution

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Busy students and professionals struggle to consistently track meals, hydration and basic health indicators. Existing solutions are often fragmented, complex or generic, making it difficult to understand daily progress and maintain healthy habits.
2	Idea / Solution Description	NutriPulse is a responsive MERN-based nutrition and health tracking application. Users can securely register, maintain a profile, log meals and nutrients, record water intake, calculate BMI, view daily dashboard summaries, analyse seven-day trends and receive rule-based personalized recommendations.
3	Novelty / Uniqueness	The application combines nutrition, hydration, BMI, analytics and profile-aware recommendations in one simple dashboard. Recommendation rules use profession, activity level, health goal and current records rather than displaying the same generic advice to every user.
4	Social Impact / Customer Satisfaction	NutriPulse encourages health awareness, hydration and consistent self-monitoring. A clean interface and immediate feedback can help users make informed lifestyle choices while the disclaimer encourages professional consultation for medical needs.
5	Business Model (Revenue Model)	The academic MVP is free. A future freemium model could offer free core tracking and paid features such as

		advanced reports, expert consultation, custom meal plans, cloud reminders and institutional wellness dashboards.
6	Scalability of the Solution	The application uses separate frontend, API and database layers with modular routes, controllers and models. It can be deployed using cloud hosting and expanded with notifications, file uploads, admin tools, mobile apps, external food APIs and advanced recommendation models.

Current MVP Scope

- Registration, login, JWT authentication and protected routes
- Profile management
- Meal Tracker CRUD with nutrition totals
- Water Tracker with daily progress
- BMI Calculator with history
- Dashboard summary and weekly analytics
- Rule-based personalized recommendations